Enhancing Hardware Trojan Detection: A Dual-Path CNN Approach to Side-Channel Analysis

Arash Golabi
Department of Computer Science and Engineering
Qatar University
Doha, Qatar
arash.golabi@qu.edu.qa

Abdelkarim Erradi
Department of Computer Science and Engineering
Qatar University
Doha, Qatar
erradi@qu.edu.qa

Ahmed Bensaid
Department of Computer Science and Engineering
Qatar University
Doha, Qatar
abensaid@qu.edu.qa

Abdulla Al-Ali
Department of Computer Science and Engineering
Qatar University
Doha, Qatar
abdulla.alali@qu.edu.qa

Uvais Qidwai
Department of Computer Science and Engineering
Qatar University
Doha, Qatar
uqidwai@qu.edu.qa

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***Abstract*—study presents a dual-path deep learning architecture designed to effectively detect and identify HT using Side-Channel Analysis (SCA). Initially, the method converts side-channel time series data—including power consumption, electromagnetic emissions, and timing information—into two separate image-like formats by applying Markov Transition Field (MTF) and reshaping techniques. This transformation effectively preserves the complex features of the data, which are crucial for detecting subtle Trojan manipulations. Subsequently, the resulting images are passed through two distinct convolutional neural network (CNN) pathways within our dual-path architecture, each optimized to enhance feature extraction. The features obtained from each pathway are then merged and input into a dense neural network layer, combining various signal characteristics to achieve robust and accurate classification. This dual-path strategy build our proposed method, enabling the simultaneous classification of both legitimate and Trojan-infected data, as well as the identification of the specific type of Trojan attack. The model has undergone extensive testing using the publicly accessible Advanced Encryption Standard (AES) dataset for HTs sourced from TrustHub and IEEE Dataport. Our method not only improves classification accuracy by using the combined strengths of CNNs but also shows marked improvements over current techniques.**

***Index Terms*—Hardware Trojan Detection, Time Series Analysis, Markov Transition Field, Convolutional Neural Networks, Deep Neural Networks, Side-Channel Signal Analysis.**

I. Introduction

In today's increasingly interconnected and technology-dependent world, electronic systems security and integrity are paramount. One of the most insidious threats to these systems is the presence of Hardware Trojans (HTs), which are malicious modifications embedded within electronic components during the design, manufacturing, or supply chain stages [1]. Unlike software-based attacks, HTs reside at the hardware level, making them particularly challenging to detect and mitigate due to their deep integration and limited visibility into hardware operations [2]. The globalization and complexity of the semiconductor supply chain have increased the risk of Hardware Trojans, allowing malicious actors to introduce subtle yet impactful alterations. In critical sectors like finance, healthcare, and national security, these Trojans can disrupt operations, leak sensitive information, and potentially cause catastrophic consequences [3].

HT attacks are highly sophisticated, often bypassing traditional testing and verification methods, which makes them a significant concern for cybersecurity experts due to their covert nature and potential for significant harm. This growing threat highlights the urgent need for robust security measures to protect hardware from such vulnerabilities and to develop next-generation defensive mechanisms [4]. Side-Channel Analysis (SCA) is a primary technique used to detect HTs by monitoring physical parameters like power consumption and electromagnetic emissions, which can reveal anomalies indicative of malicious alterations. However, SCA methods can be challenged by noise and normal operational variations, potentially leading to false positives or negatives [5].

Integrating machine learning methods with SCA significantly enhances the detection of HTs [6]. In this hybrid method, machine learning algorithms are utilized to process and examine the side-channel signals emitted by electronic devices. These algorithms develop advanced mathematical models that represent the typical physical and operational behaviors of the devices. Such modeling allows for the ef-

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fective identification and classification of irregularities and anomalies in the side-channel signals, which often indicate the presence of HTs. Various machine learning techniques, including both supervised and unsupervised methods, have been successfully applied to detect HT attacks by analyzing side-channel signals [7]. Various machine learning techniques, including both supervised and unsupervised methods, have been successfully applied to detect HT attacks by analyzing side-channel signals [7]. Techniques such as Support Vector Machine (SVM) [8], Naïve Bayes [9], Random Forest [10], and XGBoost classifiers [11] can be used to identify subtle anomalies in power traces, electromagnetic emanations, timing characteristics, and other physical observables introduced by malicious hardware modifications. Moreover, innovative techniques like Deep Belief Networks (DBN) optimized with Stacked Autoencoder Hashing Optimization (SAEHO) [12], deep transfer learning [13], and attention mechanism-enhanced models [14], represent state-of-the-art approaches that enhance detection accuracy and adaptability in the field of HT attack identification using SCA. The combination of SCA and CNN enhances feature extraction and offers robustness against noise, making it an effective tool for detecting HTs and other security threats. By converting time-series side-channel signals into image-based representations and using CNNs, this method efficiently handles high-dimensional data and reduces the need for manual feature engineering, providing a scalable and effective solution [15], [16]. These algorithms represent the advanced application of machine learning methods in HT detection to address the evolving challenges of modern cybersecurity threats.

Unlike 1D CNNs that directly process time series data, 2D CNNs require transforming this data into image-like formats to leverage their spatial pattern recognition capabilities. The Markov Transition Field (MTF) technique converts time series data into matrix representations, effectively capturing the transition probabilities between different states and preserving dynamic characteristics essential for accurate HT detection [17], [18]. This transformation enhances CNNs' ability to identify subtle anomalies and hidden dependencies that are often less discernible in the original time series format. By integrating MTF with CNNs, the detection of HTs is significantly improved through the identification of complex patterns and discrepancies in the transformed data, thereby increasing hardware security in the HT detection domain.

The primary contribution of this study is the development of a single-stage deep learning architecture for detecting and identifying HTs using SCA and time-series imaging. To the best of our knowledge, the conversion of side-channel time series data into image-like representations using MTF has never been applied in the literature for HT detection. These image formats are processed through two separate CNN pathways within a dual-path architecture, optimized for feature extraction. This framework forms the foundation of the proposed deep learning model, which simultaneously detects attacks and identifies the type of Trojan present. The effectiveness of the method is demonstrated through extensive simulations using publicly available datasets, showcasing superior performance compared to existing approaches.

II. Overview of Hardware Trojans

The importance of understanding HT attacks lies in their serious threat to the security and reliability of electronic systems, including critical infrastructure and consumer electronics, leading to serious risks that may result in catastrophic failures and breaches of privacy.

HTs can be categorized based on their activation triggers and malicious payloads [19]. The trigger mechanism refers to the specific condition or set of conditions that activate the HT. These triggers are often meticulously designed to remain concealed and undetected until a particular event occurs. They can range from simple occurrences, such as a designated date or time, to complex activity patterns and specific inputs that may happen only under rare circumstances. This advanced triggering mechanism allows the Trojan to evade detection during standard testing and normal operation, ensuring it stays dormant until it can execute its intended purpose. Once activated, the HT's payload represents the harmful action carried out by the Trojan. Depending on the payload, a hardware Trojan can modify device functionality, exfiltrate sensitive information, degrade system performance, or cause physical damage. HTs can be inserted at various stages of a chip's lifecycle, from the initial design phase to fabrication and final deployment . During the design phase, third-party Intellectual Property (IP) cores, frequently used to reduce development time and costs, can be compromised to introduce HTs covertly. Design software tools are also at risk, where malicious code or circuitry can be embedded and remain dormant until triggered by specific conditions. The fabrication stage, especially when outsourced to untrusted facilities, is another vulnerable point, where chip layouts can be modified, or lithography masks tampered with to embed HTs directly into the silicon. Although global outsourcing is economically advantageous, the complexity and microscopic scale of modern integrated circuits make such alterations nearly undetectable. Additional vulnerabilities exist during the assembly and packaging stages, where HTs can be inserted by modifying the silicon die or embedding malicious components within the chip package. Furthermore, as chips move through the supply chain, they are exposed to the risk of interception and tampering, adding yet another layer of vulnerability.

Given the complexity of these threats, robust security measures must be implemented throughout a chip's entire lifecycle. A holistic approach is essential to mitigate the risks posed by HTs and to safeguard the security and functionality of electronic systems. This approach should encompass secure and verified design practices, integrity checks during manufacturing and assembly, and supply chain security. Additionally, the development and deployment of advanced detection techniques are crucial for identifying and thwarting Trojans that bypass initial preventative measures. Without an effective HT detection mechanism, critical electronic infrastructures remain vulnerable to significant compromise.

III. Deep learning method for hardware Trojan detection

This section outlines the proposed methodology for detecting and identifying HT threats. Time series data is transformed using MTF and reshaping techniques into two 2D images, allowing CNNs to apply their spatial feature learning capabilities. The extracted features are then classified to detect the presence of an HT attack and identify the specific Trojan type.

A. Transforming time series data into images for analysis

Converting time series data into image-like representations has demonstrated substantial performance enhancements across various domains, allowing CNNs to be effectively applied for time series analysis by using their ability to recognize spatial features [20], [21].

One approach to transforming time series data into an image-like format is the reshaping technique, where the linear sequence of data points is arranged into rows and columns to form a 2D matrix [22]. This method organizes the data row-wise into specified dimensions while maintaining the sequential order of the time series. Another innovative technique is MTF imaging, which transforms time series data into a matrix representing transition probabilities, capturing the underlying dynamics and dependencies between different states in the data [23]. This matrix illustrates the probability of transitioning between different quantized states, offering a detailed visual representation of the temporal dynamics within the time series [24]. Converting time series data into a transition probability matrix enables the visualization and analysis of complex temporal patterns in a two-dimensional space. The MTF technique is especially beneficial as it encodes both temporal variations and state dependencies into spatial patterns, making it ideal for use with two-dimensional CNNs. These CNNs are highly effective at identifying spatial relationships and patterns within the data, improving the model's capacity to detect hidden features and intricate patterns [18].

1) Implementing MTF for Time Series Data Analysis: MTF is a method for transforming time series data into a matrix format suitable for image-based analysis, such as with CNNs. It represents the time series as a transition probability matrix, effectively encoding temporal dynamics into a spatial representation.

First, consider the time series data $\alpha_t = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$. The data is normalized to scale the values between $[-1, 1]$:

$$\tilde{\alpha}_t = \frac{2(\alpha_t - \min(\alpha_t))}{\max(\alpha_t) - \min(\alpha_t)} - 1 \tag{1}$$

After normalization, the time series is quantized into k discrete levels by mapping the normalized values to one of k bins:

$$\hat{\alpha}_t = \left\lfloor k \times \frac{\tilde{\alpha}_t - \min(\tilde{\alpha}_t)}{\max(\tilde{\alpha}_t) - \min(\tilde{\alpha}_t)} \right\rfloor \tag{2}$$

Here, k is the number of quantization levels, and $\hat{\alpha}_t$ represents the quantized version of the time series.

Next, compute the transition probabilities. For each pair of quantized states (i, j), calculate the probability of transitioning from state i to state j:

$$P_{ij} = \frac{\sum_{t=1}^{n-1} \mathbb{1}(\hat{\alpha}_t = i \wedge \hat{\alpha}_{t+1} = j)}{\sum_{t=1}^{n-1} \mathbb{1}(\hat{\alpha}_t = i)} \tag{3}$$

where $\mathbb{1}$ is the indicator function, which is 1 if the condition is true and 0 otherwise.

The MTF is constructed as a $k \times k$ matrix where each element M_{ij} represents the transition probability P_{ij}:

$$M_{ij} = P_{ij} \tag{4}$$

The resulting MTF matrix M can be visualized as an image, with each pixel intensity corresponding to the transition probability between states. This transformation encodes the dynamical relationships and dependencies between different states of the time series into a two-dimensional spatial format. This format is suitable for processing by CNNs for tasks such as classification, regression, or anomaly detection. By following these steps, the time series data is transformed into an image-like representation, enabling the application of advanced image processing techniques for various analytical tasks.

It is important to highlight that adjusting the size of the k-sized time-series data windows can be achieved through various methods. A commonly used approach is resampling, which involves downsampling to reduce the size or upsampling to increase it. To modify the dimensions of the output image, the quantization levels are first adjusted to correspond with the target image size.

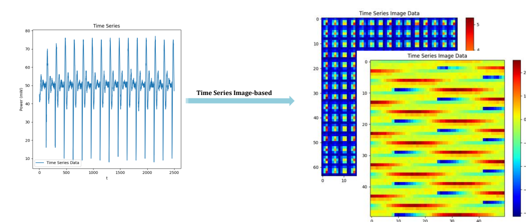

Fig. 1. Visualizing Time Series with MTF and Reshaping: (Left) Original Time Series, (Right) MTF and reshaping Representation.

Figure 1 shows the result of transforming a sample time-series data sample using the reshaping and MTF method into an image-like matrix suitable for use by CNNs.

B. One-Stage deep learning framework for HT detection and identification

The images obtained by transformation in previous sections then undergo processing through a series of CNN blocks in a two-path structure within the model's architecture, as presented in Figure 2. Each CNN block applies filters to extract features, enhanced by batch normalization for stable and efficient training. A ReLU function introduces non-linearity, allowing the model to capture complex patterns. After convolution, a pooling layer downsizes the data, retaining key

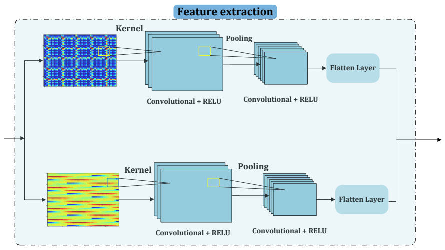

Fig. 2. The feature extraction block using two 2D image representations from time series data

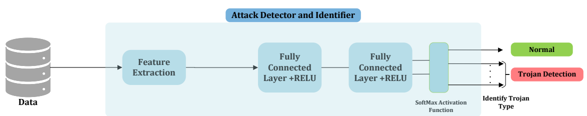

Fig. 3. One stage Attack Detector and Identifier.

features and reducing computational load. This combination of CNN and pooling layers is repeated to ensure thorough feature extraction. The output from each path is flattened into one-dimensional vectors and concatenated, then passed through fully connected layers. Finally, a Softmax classifier predicts whether an HT attack is triggered.

The feature extraction module, integrated with the classifier component, constitutes the model for the attack detector and identifier part as shown in Figure 3. The proposed model operates as a two-stage framework specifically designed to enhance HT attack detection and classification. In the first stage, named the Attack Detector, the model processes the transformed time-series data to determine whether it indicates a normal state or suggests the presence of a HT attack. Upon detecting an HT attack, the model activates the second stage, the Attack Identifier. This stage is tasked with classifying the specific type of Trojan attack identified initially. This two-stage approach not only ensures the accurate detection of potential security breaches but also enables a detailed classification of the specific type of threat, thereby facilitating more targeted and effective countermeasures. The model's architecture is designed to integrate these stages seamlessly, ensuring efficient and robust performance in the identification and categorization of hardware Trojan attacks.

C. Overview of Dataset

The dataset used for this research was obtained from IEEE DataPort and contains power side-channel signals from a subset of HTs designed to target an AES-128 cryptographic core, as referenced in the Trust-Hub repository [25], [26]. The AES circuit encrypts 128-bit plaintext using a secret key, and data was collected over 10,000 encryption processes, generating time-series signals of $2,500$ points each. The data was gathered using Sakura-G boards, capturing power consumption under various conditions, including HT activation and inactivity, offering a comprehensive basis for HT detection models [26].

This study focuses on five specific Trojan types, namely AES-T500, AES-T600, AES-T700, AES-T800, and AES-T1600, for comparison with previous works. These Trojans vary in their activation methods and impact on the AES circuit, such as leaking the secret key through covert channels or RF signals. For training and validation, 80% of the signals from both the disabled and triggered Trojan sets were used for training, while 20% were set aside for validation, ensuring robust model development.

IV. Results and Discussions

In this section, results and discussion are provided about the novel time series imaging-based HT detection method to evaluate the performance of proposed method.

A. Evaluation of the proposed HT detection method

For the purpose of simulation and evaluation of our proposed time series imaging-based learning method, a selection is made where 80% signals are allocated for the training set and 20% for the validation set. In our study, we explore the application of the model, as depicted in Figure 3 and implemented using PyTorch version 2.1.2, to accurately detect HT attacks.

The model begins with two convolutional layers for feature extraction. The first layer uses 32 filters with a kernel size of 5, stride of 1, and padding of 2 to capture initial features, followed by a max pooling layer (kernel size 2, stride 2) to reduce spatial dimensions. The second convolutional layer applies 64 filters with the same configuration to refine the extracted features. After feature extraction, the classification phase starts with fully connected layers. The layers first map the features into a 128-dimensional space, and the final output layer produces a 6-dimensional classification result, identifying either a "normal" condition or one of the five Trojan attack types. Table I presents the CNN model architecture parameters.

TABLE I
CNN Model Architecture for Attack Detector and Attack Identifier

Layer Type	Filters/Nodes	Kernel Size	Stride	Padding
Convolutional Layer 1	32	5	1	2
Batch Normalization Layer 1	32	-	-	-
Max Pooling Layer	-	2	2	-
Convolutional Layer 2	64	5	1	2
Batch Normalization Layer 2	64	-	-	-
Max Pooling Layer	-	2	2	-
Fully Connected Layer 1	128	-	-	-
Fully Connected Layer 2	6	-	-	-

In this study, we employ the cross-entropy loss function for both binary and multi-class classification tasks. The binary classification loss is expressed as follows:

$$L(y, \hat{y}) = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)\right] \quad (5)$$

where y represents the true label, $\hat{y}$ the predicted probability of the positive class, and N the number of samples in the dataset.

Here, y_i indicates the presence or absence of an HT, and $\hat{y}_i$ is the predicted probability that the i-th sample contains an HT.

For this problem, which involves multi-class classification, the loss function is adapted to cater to multiple classes and is defined as:

$$L(y, \hat{y}) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} [y_{i,c} \cdot \log(\hat{y}_{i,c})] \tag{6}$$

where C is the number of classes, $y_{i,c}$ is a binary indicator of whether class label c is the correct classification for observation i, and $\hat{y}_{i,c}$ is the predicted probability of observation i being of class c. Accuracy for both binary and multi-class scenarios is calculated by:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}(\hat{y}_i = y_i) \tag{7}$$

where $\mathbb{1}$ (.) is the indicator function that is 1 when $\hat{y}_i$ equals y_i and 0 otherwise.

To evaluate the effectiveness of the proposed HT detection methods, a comprehensive comparison is conducted with several established models. This includes the One-Class Support Vector Machine (OC-SVM), known for its strength in anomaly detection, as well as three neural network architectures: MLP, CNN, and Hierarchical Temporal Memory (HTM), the latter being inspired by the human neocortex for processing temporal data. The performance metrics for these models are sourced from the results reported in [26]. Additionally, the comparison extends to the Attention-Res-Attention (ARA) network from [15], which integrates the ResNeXt architecture with an attention mechanism to enhance the detection of critical HT-related features.

To establish a comprehensive comparative analysis, we trained and evaluated the proposed Attack Detector and Identifier model in Figure 3, on the AES datasets. The performance of the proposed model in detecting HTs in AES encryption modules is demonstrated through the overall accuracy and loss, as shown in the Figure 5. The results of this evaluation for each AES dataset are presented in Table II, which provides a detailed comparison of the detection rates achieved by the model for each AES dataset. It should be noted that to obtain the accuracy for each class, we first filter the validation data to include both normal samples and samples from the specific class being evaluated. The model is then evaluated on this balanced subset, where the number of correct predictions for both normal and attack samples is tracked. Class-specific accuracy is computed as the ratio of correctly predicted samples to the total samples within that class, with separate metrics for attack and normal accuracy. As evidenced by the table, our proposed time-series imaging-based detection approach exhibits superior performance on the AES-600 and AES-1600 hardware Trojan benchmarks, while maintaining comparable performance to existing methods on the AES-500, AES-700, and AES-800 benchmarks.

The proposed model achieves an accuracy of approximately 93.25% on the AES-600 benchmark and 75.40% on the AES-T1600 benchmark, each exceeding the respective accuracy levels reported in previous studies [15], [26]. These results emphasize the efficacy of our approach in accurately detecting hardware Trojan attacks, particularly in scenarios where existing methods may not be effective. The comprehensive analysis presented in Table II substantiates the competitive performance of our proposed methodology, demonstrating its potential to advance the state-of-the-art in hardware Trojan detection and identification. The AES-T1600 dataset is particularly challenging compared to other datasets for HT detection due to its complex signal patterns and subtle variations when the Trojan is activated as shown in Figure 4. Despite these difficulties, our proposed method achieves an accuracy of 75.40%, which is a notable accuracy compared to the reported accuracy of 59.6% in [26] and 63.1% in [15]. This result demonstrates the effectiveness of the approach in addressing HT detection difficulties in challenging datasets. Table III presents the Recall, Precision, and F1-Score metrics across five AES benchmarks. The model shows excellent performance, achieving an F1-Score of 1.00 for AES-T700 and AES-T800, indicating perfect detection with no false positives or negatives. However, in more challenging datasets like AES-T600 and AES-T1600, while Recall and Precision remain strong, F1-Scores drop slightly to 0.9325 and 0.7537, respectively.

TABLE II
COMPARISON OF ACCURACY OF ATTACK DETECTOR WITH DIFFERENT METHODS ON EACH AES DATASETS

	AES-T500	AES-T600	AES-T700	AES-T800	AES-T1600
CNN	84.5%	55.5%	100%	100%	64%
MLP	84%	65.5%	100%	100%	70.5%
OC-SVM	89.7%	49.7%	90.2%	92.7%	50.2%
HTM [26]	100%	63.5%	98.1%	100%	59.6%
ARA [15]	100%	67.3%	100%	100%	63.1%
Our Method	98.5%	93.25%	100%	100%	75.40%

TABLE III
PERFORMANCE METRICS OF THE ATTACK DETECTION MODEL ACROSS AES BENCHMARKS

	AES-T500	AES-T600	AES-T700	AES-T800	AES-T1600
Recall	0.9855	0.9325	1	1	0.755
Precision	0.9857	0.9327	1	1	0.76
F1-Score	0.9855	0.9325	1	1	0.7537

V. CONCLUSION

This paper has presented a novel approach for detecting Hardware Trojans (HTs) in chip devices, utilizing a combination of Markov Transition Field (MTF) transformations and deep learning techniques. By transforming side-channel signals into image-like formats through MTF, we utilize the convolutional neural networks (CNNs) to effectively address the intricate task of HT detection. Our evaluation, using the Advanced Encryption Standard (AES) dataset from TrustHub and IEEE Dataport, has demonstrated the superior accuracy and classification precision of our method over existing techniques. The proposed approach, characterized by its robust feature extraction and generalization capabilities, achieved an accuracy of approximately 93.25% for AES-600 and 75.40% for AES-T1600, surpassing previous benchmarks.

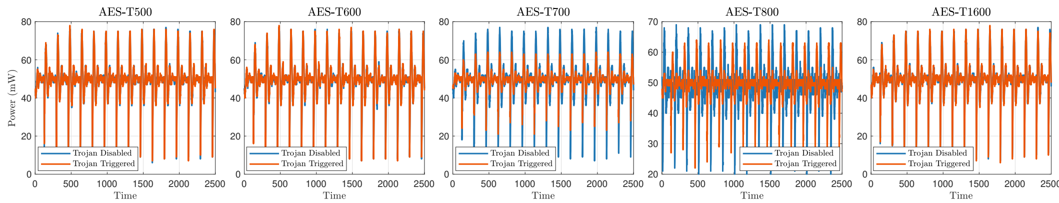

Fig. 4. Comparison of Trojan Disabled vs. Trojan Triggered for various datasets.

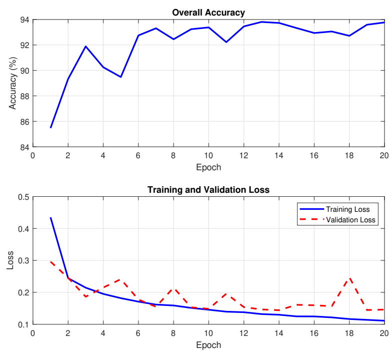

Fig. 5. Overall accuracy of the model across training epochs

Future work will focus on designing algorithms that are more robust to noise and developing methods that can generalize across a wider range of dataset, enabling the identification of different types of HT attacks with increased accuracy.

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